ENTERPRISE MACHINE LEARNING • CUSTOMER RETENTION

Telecom Customer Churn Prediction Engine

Proactive retention analytics powered by production-grade machine learning

Presenter: [Your Name]



BASE-RATE SPLIT	CHURN RECALL	INFERENCE TARGET
73.4 / 26.6	~82%	<20 ms

PREDICT • PRIORITIZE • RETAIN

Why churn is an existential risk

Reactive retention loses customers before teams can intervene.

The operating goal is not simply to predict churn. It is to identify customers whose retention is both achievable and economically worthwhile.

EXPECTED VALUE OF AN INTERVENTION

P(Churn) × [P(Accept) × (CLV saved – Offer cost)]

FALSE NEGATIVE

Miss a churner → lost lifetime value

FALSE POSITIVE

Discount loyal customer → margin cost

Decision principle: choose the threshold using intervention capacity and business costs—not accuracy alone.

What drives customers to leave?

CONTRACT TERM

Month-to-month: 42.7% churn
Two-year: 2.8% churn

Long-term commitment is a strong retention signal.

EDA SIGNAL

SERVICE & SUPPORT

Fiber optic: 41.9% churn
DSL: 19.0% churn

Review support and security coverage for fiber customers.

EDA SIGNAL

PAYMENT FRICTION

Electronic check: 45.3% churn
Auto-pay: ~15.5% churn

Reduce billing friction and encourage convenient payment.

EDA SIGNAL

Data hygiene & feature engineering

01

Clean

Convert TotalCharges to numeric; handle 11 blank-string records associated with tenure = 0.

02

Engineer

Create six tenure lifecycle cohorts to capture non-linear onboarding and retention patterns.

03

Encode

Apply one-hot encoding with drop_first=True to reduce redundant dummy columns.

04

Align & scale

Align inference inputs to the 34-column template and apply StandardScaler.

Production rule: fit preprocessing on training data only, then reuse the exact fitted transforms at inference.

The class-imbalance bake-off

A model predicting “No Churn” for everyone can score 73.4% accuracy while detecting 0% of churners.

Strategy	F1-score	Recall	Production trade-off
Unweighted baseline	0.53	0.49	Simple; weak minority capture
SMOTE	0.60	0.74	Synthetic samples; added memory
SMOTEENN	0.61	0.79	Noise cleaning; slower training
Cost-sensitive weighting	**0.64**	**0.82**	**No synthetic rows; native support**

Reported best: sample weighting (positive-class weight ≈ 2.76). Validate against the actual deployment cost curve.

Benchmark models, then optimize

MODELS BENCHMARKED

- Decision Tree
- Random Forest
- AdaBoost
- CatBoost
- LightGBM
- XGBoost

OPTUNA SEARCH

100 trials • 5-fold stratified CV • minority-class F1 objective

Search space: learning rate, depth, subsampling, regularization and class weighting

TUNED XGBOOST

F1 0.634 | Recall 0.79

COST-WEIGHTED ADABOOST

F1 0.641 | Recall 0.82

Early stopping: validation log-loss plateau for 15 iterations (as specified).

Why AdaBoost was selected

01

Metric parity

Reported F1 ≈ 0.64 and recall ≈ 0.82, competitive with more complex boosters.

02

Operational simplicity

Scikit-learn-native deployment reduces dependency and packaging complexity.

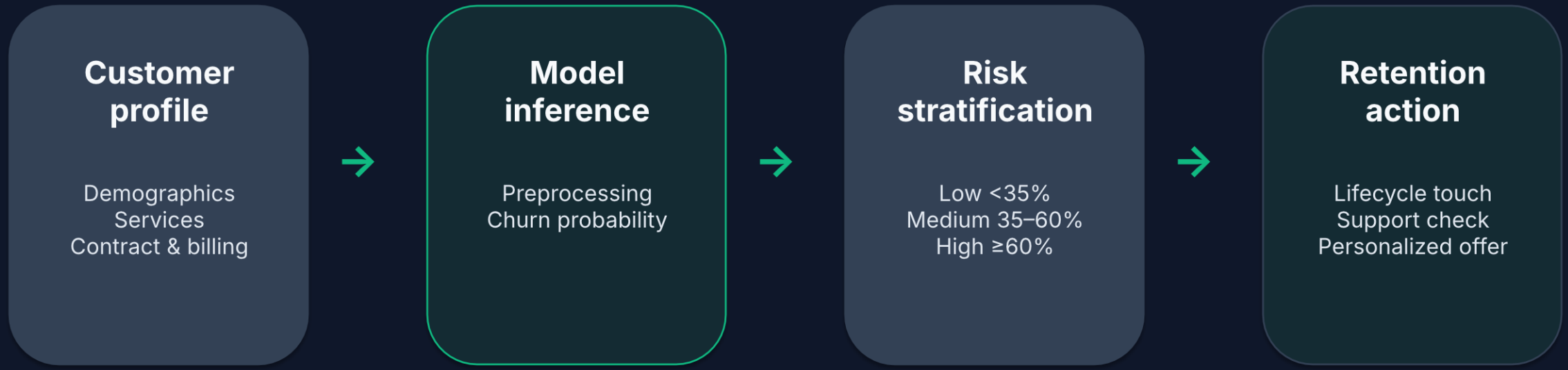
03

Low serving overhead

Shallow tree ensembles are typically fast; benchmark latency in the target container.

Selection principle: choose the model that meets business targets with the lowest sustainable operational risk.

From prediction to intervention



Explainability card example: “Month-to-month contract + fiber service + no tech support.”

Risk bands are illustrative operating thresholds; tune them to campaign capacity and validated calibration.

Containerization & security architecture

NON-ROOT

Run as dedicated appuser (UID 10001).

SLIM BASE

Use python:3.11-slim-bookworm where dependencies allow.

LAYER CACHE

Install dependencies before copying frequently changed source.

HEALTH CHECK

Probe Streamlit’s /_stcore/health endpoint.

Also add .dockerignore, pin dependencies, scan images, and avoid baking secrets into the image.

Multi-cloud deployment options

STREAMLIT COMMUNITY CLOUD

FASTEST PATH

GitHub-linked deployment; managed hosting and secrets configuration.

AWS EC2 + NGINX

MORE CONTROL

VM-based serving with reverse proxy, process supervision and TLS.

AWS ECS FARGATE + ALB

MANAGED SCALE

Container tasks behind an Application Load Balancer; configure scaling and health checks.

Choose the deployment target based on reliability, traffic, security requirements, and operating budget.

Observability & drift detection

MONITOR INPUT DISTRIBUTIONS

Population Stability Index (PSI) Wasserstein distance

A PSI threshold above 0.25 is a proposed alert rule in the source prompt—not a universal standard. Calibrate it to baseline variation.

HANDLE DELAYED LABELS

Churn labels may arrive 30–90 days later

Track performance once outcomes mature. Use drift as an early warning, not proof that model quality has degraded.

Experiment design: keep a randomized 5–10% control group to estimate incremental retention lift.

Business impact & engineering roadmap

CHURN RECALL	ARR PROTECTED	POPULATION
~82%	$420k+*	50,000*

**Illustrative estimates from the supplied prompt; validate with actual unit economics and experiment results.*

Q1 Production baseline

Serialize the full preprocessing + model pipeline; automate CI tests.

Q2 Serving & registry

Move UI/API boundary to FastAPI; evaluate ONNX Runtime and add MLflow registry.

Q3 Causal targeting

Explore uplift modeling to target customers most likely to respond to offers.

Turn billing signals into measurable retention outcomes—with controlled experiments, not prediction scores alone.